

Bootstrap

Button :-

- A button is an interactive element used to perform actions like submitting forms, navigation, or triggering events.
- In Bootstrap, buttons are styled using predefined classes, so you don't need to write custom CSS.
- Every button must include the base class btn to apply Bootstrap's button styling.



```
<button type="button" class="btn btn-primary">Primary</button>
```



Example –

```
<!--Button-->
<button class="btn btn-primary">Click</button>
<button class="btn btn-outline-primary">Click</button>
<button class="btn btn-secondary">Click</button>
<button class="btn btn-outline-secondary">Click</button>
<button class="btn btn-success">Click</button>
<button class="btn btn-outline-success">Click</button>
<button class="btn btn-warning">Click</button>
<button class="btn btn-outline-warning">Click</button>
<button class="btn btn-dark">Click</button>
<button class="btn btn-outline-dark">Click</button>
<button class="btn btn-light">Click</button>
<button class="btn btn-outline-light">Click</button>
```



Table :-

- A table is used to display structured data in rows and columns.
- In Bootstrap, the base class for tables is table.
- Bootstrap provides extra classes to style tables with different looks:
 - table-striped → Adds zebra-striping to rows.
 - table-bordered → Adds borders to all cells.
 - table-hover → Highlights row when hovered.
 - table-dark → Dark themed table.
 - table-borderless → Removes borders.
 - table-sm → Makes table more compact (small).
- Tables in Bootstrap are also responsive using the class table-responsive.



The **.table** class is used to add basic styling to a table.

```
<table class="table">
  <tr>
    <th>Id</th>
    <th>Name</th>
  </tr>
  <tr>
    <td>101</td>
    <td>Abc</td>
  </tr>
  <tr>
    <td>102</td>
    <td>Xyz</td>
  </tr>
</table>
```

table table-striped

table table-bordered

table table-hover

Example –

```
<!-- Table -->
<table class="table table-hover table-dark">
  <tr class="table-primary">
    <th>ID</th>
    <th>NAME</th>
    <th>CITY</th>
  </tr>

  <tr class="table-danger">
    <td>1</td>
    <td>A</td>
    <td>Pune</td>
  </tr>

  <tr class="table-success">
    <td>2</td>
    <td>B</td>
    <td>Satara</td>
  </tr>

  <tr class="table-warning">
    <td>3</td>
    <td>C</td>
    <td>Mumbai</td>
  </tr>
</table>
```

ID	NAME	CITY
1	A	Pune
2	B	Satara
3	C	Mumbai

Form :-

- A form is used to collect user input (like text, email, password, etc.).
- In Bootstrap, the base class for forms is form-control (applied to input, select, textarea).
- Forms in Bootstrap are responsive, neatly styled, and mobile-friendly without custom CSS.

Important classes in Bootstrap forms:

- form-label - Styles labels.
- form-control - Styles inputs, textareas, selects.
- form-check - Styles checkboxes and radio buttons.
- form-check-label - Styles labels of checkboxes/radios.
- form-select - Styles dropdowns.
- btn - Styles submit/reset buttons.

Form

Bootstrap Form Rules

- Always use `<form role="form">` (helps improve accessibility for people using screen readers)
- Wrap labels and form controls in `<div class="form-group">` (needed for optimum spacing)
- Add class `.form-control` to all textual `<input>`, `<textarea>`, and `<select>` elements

Form

```
<div class="form-group">
  <label for="inputEmail1">Email address</label>
  <input type="email" class="form-control" id="inputEmail1" placeholder="Email">
</div>

<div class="form-group">
  <label for="InputPassword">Password</label>
  <input type="password" class="form-control" id="InputPassword" placeholder="Password">
</div>

<button type="submit" class="btn btn-default">Login</button>
```

Example -

```
<!--Form-->
<div class="mb-4">
  <label for="UserID" class="form-label">UserID:</label>
  <input type="text" class="form-control-lg">
</div>

<div class="mb-4">
  <label for="Password" class="form-label">Password:</label>
  <input type="password" class="form-control-lg">
</div>

<button class="btn btn-outline-dark">Submit</button>
<br><br>
```

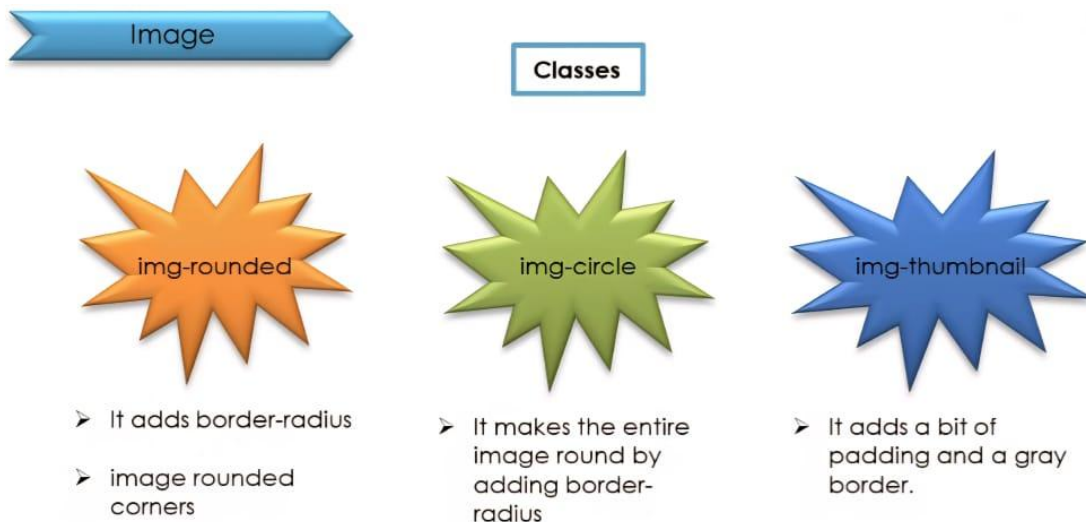
UserID:

Password:

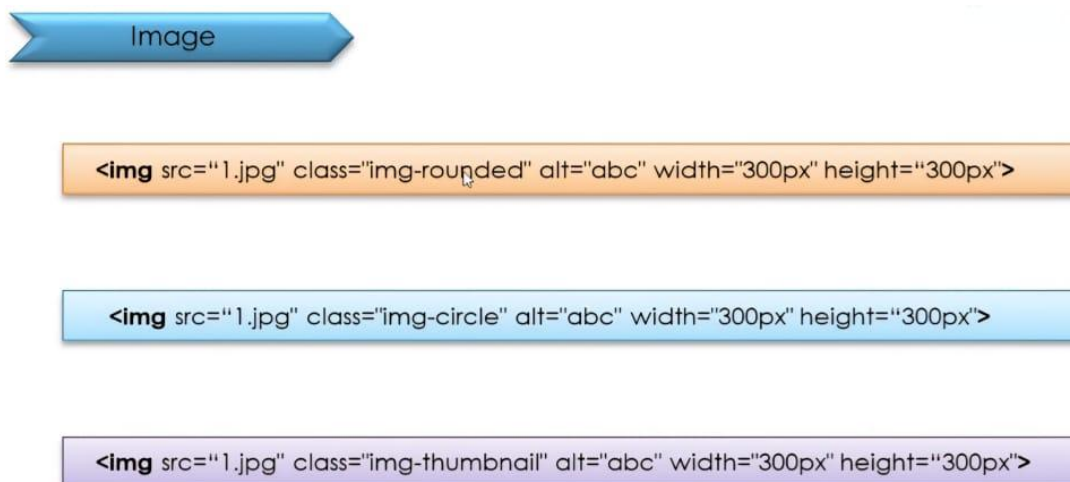
Submit

Image :-

- Images in Bootstrap are made responsive and styled using utility classes.
- The most common class is `img-fluid`:
 - Makes the image responsive (scales with parent container).
 - Applies `max-width: 100%` and `height: auto`.



- Bootstrap provides additional image classes:
 - `rounded` → Adds rounded corners.
 - `rounded-circle` → Makes the image circular.
 - `img-thumbnail` → Adds a border + padding around the image (like a photo frame).
 - `float-start` → Floats image to the left.
 - `float-end` → Floats image to the right.
 - `mx-auto d-block` → Centers the image.



```

```

```

```

```

```

- Images can also be placed inside cards, grids, or responsive containers.

Example –

```
<!--img-->
```

```
  
  
  

```



Spinner :-

- A spinner in Bootstrap is used to show a loading indicator (when something is processing).
- The base class for spinners is spinner-border (for a circular border animation).
- Another type is spinner-grow (pulsating dots).

You can change colors using text color utility classes:

- text-primary → Blue spinner
- text-success → Green spinner
- text-danger → Red spinner
- text-warning → Yellow spinner
- text-info → Light blue spinner
- text-dark, text-light → Dark/Light spinner.

You can change sizes:

- spinner-border-sm or spinner-grow-sm → Small spinner.
- Default → Normal size

Example -

```
<!--Spinner-->
```

```
<div class="spinner-grow text-muted"></div><span>Loading..</span>
```

```
<div class="spinner-border text-primary"></div><span>Loading..</span>
```



Alert :-

- An alert is used to display messages, warnings, errors, or success notifications to the user.
- In Bootstrap, alerts are created using the class alert.

Bootstrap provides **contextual classes** to show different types of alerts:

- alert-primary → Blue alert (general info).
 - alert-secondary → Grey alert.
 - alert-success → Green alert (success message).
 - alert-danger → Red alert (error or danger).
 - alert-warning → Yellow alert (warning message).
 - alert-info → Light blue alert (info message).
 - alert-light → White alert.
 - alert-dark → Black alert.
- Alerts can also be **dismissible** (closeable) using the class alert-dismissible and a close button (btn-close).



- Bootstrap Alerts are used to provide an easy way to create predefined alert messages. Alert adds a style to your messages to make it more appealing to the users.
- There are four classes that are used within <div> element for alerts.



```
<div class="alert alert-success">  
  <strong>Success!</strong> Registration Successfully Completed  
</div>
```

Closing Alert :-

- To make an alert closable, you must add:
 - The class alert-dismissible → enables close button styling.
 - The class fade show → adds smooth animation when closing.
- A close button is created using:
 - <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
 - btn-close → Bootstrap's built-in close (x) icon.
 - data-bs-dismiss="alert" → tells Bootstrap to dismiss (close) the alert when clicked.

- Bootstrap requires JavaScript bundle (bootstrap.bundle.min.js) for the closing functionality.

Closing Alert

- If you want to close the alert message, you have to add an .alert-dismissible class to the alert container.

```
<div class="alert alert-success alert-dismissible">  
  <button type="button" class="close" data-dismiss="alert">×</button>  
  <strong>Success!</strong> Registration Successfully Completed  
</div>
```

Example -

```
<!--Alert-->  
  
<div class="alert alert-success alert-dismissible">  
  <button class="btn-close" data-bs-dismiss="alert"></button>  
  <b>Registration Successfully!!</b>  
</div>
```

× Registration Successfully!!